Cheng-Hsiang Chiu

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Qualification

- Ph.D. with expertise in parallel and heterogeneous computing and task-graph scheduling
- Programming skills in C++, CUDA, and parallel frameworks with a focus on high-performance system design
- Proven ability to collaborate with cross-disciplinary teams and translate research into efficient solutions

Education

- **University of Wisconsin-Madison** — Madison, Wisconsin, USA
  Ph.D. in Electrical and Computer Engineering — *Aug. 2023 – Aug. 2025*
- **École Polytechnique Fédérale de Lausanne** — Lausanne, Switzerland
  M.S. in Computer Science — *Sep. 2013 – Feb. 2016*
- **National Chiao Tung University** — Hsinchu, Taiwan
  M.S. in Communications Engineering — *Sep. 2005 – Aug. 2007*
- **National Chung Cheng University** — Chiayi, Taiwan
  B.S. in Electrical Engineering — *Sep. 2001 – Jun. 2005*

Research

- **High Performance Computing**

  Enhanced Taskflow's parallel execution capabilities by designing and implementing new programming models for high-performance applications

  - Implemented Pipeflow to support pipeline parallelism, improving throughput in staged computational workflows
  - Built AsyncTask to overlap task creation and execution, reducing runtime overhead in dynamic task graphs
  - Built syclFlow, enabling developers to express SYCL applications as task graphs and offload computation to GPUs for acceleration
  - Conducted thorough profiling and performance tuning to identify and eliminate bottlenecks
  - Developed unit tests, documentation, and concurrency primitives, including thread pools, synchronization mechanisms, and multi-threaded task scheduling

- **Reinforcement Learning-based Scheduling**

  Developed reinforcement learning (RL)–based scheduling algorithms to improve resource efficiency and execution performance across task graphs, heterogeneous runtimes, GPU execution, and EDA workflows

  - Designed a resource-efficient task scheduling system that dynamically assigns tasks to underlying computing units to maximize utilization and reduce execution overhead
  - Designed a RL–generated topological ordering for dynamic task graph scheduling, improving execution efficiency over a work-stealing runtime
  - Developed a RL-based optimization framework for CUDA Graph scheduling, automatically inserting a limited number of dependencies to improve resource utilization in the CUDA Graph runtime

- Enhanced graph partitioning for EDA applications using RL–based initialization, providing higher-quality starting solutions for downstream optimization

Work Experience

- **University of Utah**, Salt Lake City, Utah, USA — Aug. 2020 - Jul. 2023

  Ph.D. Student

  Enhanced Taskflow's parallel execution capabilities by designing and implementing new programming models for high-performance applications

  - Implemented Pipeflow to support pipeline parallelism, improving throughput in staged computational workflows
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  - Built syclFlow, enabling developers to express SYCL applications as task graphs and offload computation to GPUs for acceleration
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- **Intel**, Austin, Texas, USA — May. 2022 - Aug. 2022

  Software Developer Intern

  Contributed to the SYCL Graph execution model, which represents a program as a Directed Acyclic Graph (DAG) of computational tasks and dependencies, enabling efficient and reusable execution on heterogeneous hardware

  - Developed core components of the SYCL Graph runtime and execution model, enabling users to define complex task graphs once and execute them efficiently across devices
  - Worked on graph construction and dependency management to ensure correct execution ordering and performance scalability

- **Cadence**, Austin, Texas, USA — May. 2021 - Aug. 2021

  Software Developer Intern

  Worked on performance-critical implementation for buffer insertion, a key physical-design optimization used to meet timing constraints, reduce interconnect delay, and preserve signal integrity in large-scale IC designs

  - Accelerated buffer insertion runtime by 16% through restructuring performance-critical code paths
  - Designed and integrated a memory pool allocator to optimize large volumes of small-object allocations, reducing allocation overhead and improving cache locality
  - Conducted thorough performance profiling using perf and gprof to identify bottlenecks and validate optimizations on Linux servers

- **The Arctic University of Norway**, Tromsø, Norway — Feb. 2019 - Dec. 2019

  Researcher

  Developed an energy-efficient, in-situ wildlife classification framework to enable continuous monitoring of Arctic animals, supporting research into the effects of climate change on ecosystems in the Arctic tundra

- Designed a low-power inference pipeline capable of operating in remote, power-constrained Arctic environments without reliance on continuous connectivity
- Implemented on-device animal classification to reduce data transmission costs and enable real-time, in-field analysis
- Optimized the system for energy efficiency and long-term deployment, balancing accuracy, latency, and power consumption
- Collaborated with domain researchers to ensure the framework met practical requirements for ecological and climate impact studies

- **Khalifa University**, Abu Dhabi, United Arab Emirates — Jan. 2018 - Nov. 2018

  Researcher

  Improved the performance and usability of python-meep for computational materials simulations by introducing parallel execution and developing visualization frameworks for simulation results

  - Parallelized python-meep workflows to accelerate large-scale electromagnetic and materials simulations
  - Integrated parallel execution to improve scalability and reduce time-to-solution for compute-intensive workloads
  - Developed visualization tools to analyze and present simulation outputs, improving result interpretation and debugging
  - Enabled more efficient experimentation and analysis for researchers working on materials and photonics simulations

- **CERN**, Genève, Switzerland — Mar. 2015 - Aug. 2015

  Software Developer

  Developed software to discover, validate, and monitor devices across CERN's large-scale data acquisition (DAQ) network, which supports hundreds of thousands of sensors used for particle detection and system monitoring

  - Built network discovery tools to automatically identify and register devices within the DAQ infrastructure
  - Implemented consistency checks between live network state and installation databases, ensuring configuration correctness and reducing operational errors
  - Developed monitoring capabilities to track device health, connectivity, and network status in real time
  - Contributed to the reliability and observability of mission-critical systems supporting high-energy physics experiments

- **National Chiao Tung University**, Hsinchu, Taiwan — Jan. 2009 - Jun. 2013

  Research Assistant

  Worked on the design and deployment of large-scale data, networking, and multimedia systems across healthcare, smart infrastructure, and assistive-technology projects

  - Built a data center infrastructure for storing and managing large volumes of medical brain imaging data for a remote healthcare initiative
  - Designed and implemented a load-balancing algorithm in a peer-to-peer network

- Constructed wireless sensor networks (WSNs) to monitor structural health, enabling continuous data collection and fault detection
- Coordinated and deployed campus-wide surveillance systems and developed assistive devices to help guide visually impaired students
- Built and deployed a multimedia streaming platform within a vision-based intelligent environment

Skills

- **Programming Language**: C, C++, Python, HTML, SQL
- **Programming Model**: Taskflow, SYCL, CUDA, OpenMP, MPI, PyTorch
- **Unit Test**: doctest
- **Profiler**: gprof, perf
- **Software Development**: GitHub, CI/CD

Publications

- **Thesis**

  - Cheng-Hsiang Chiu, "Asynchronous Many-task System with Intelligent Scheduling," Ph.D. Dissertation, Department of Electrical and Computer Engineering, University of Wisconsin-Madison, 2025
  - Cheng-Hsiang Chiu, "Protection against Data Profiling by Adversarial Cloud Applications," M.S. Thesis, Department of Computer Science, École Polytechnique Fédérale de Lausanne, 2016
  - Cheng-Hsiang Chiu, "Process Control in Streaming Server," M.S. Thesis, Department of Communication Engineering, National Chiao Tung University, 2007

- **Journal Papers**

  - Wan-Luan Lee, Dian-Lun Lin, Shui Jiang, **Cheng-Hsiang Chiu**, Yibo Lin, Bei Yu, Tsung-Yi Ho, and Tsung-Wei Huang, "G-kway: Multilevel GPU-Accelerated k-way Graph Partitioner using Task Graph Parallelism," *ACM Transactions on Design Automation of Electronic Systems (TODAES)*, 2025
  - Yu-Cheng Chiou, Tuza Adeyemi Olukan, Mariam Ali Almahri, Harry Apostoleris, **Cheng-Hsiang Chiu**, Chia-Yun Lai, Jin-You Lu, Sergio Santos, Ibraheem Almansouri, and Matteo Chiesa, "Direct Measurement of the Magnitude of van der Waals interaction of Single and Multilayer Graphene," *ACS LANGMUIR*, 2018

- **Conference Papers**

  - Chedi Morchdi, **Cheng-Hsiang Chiu**, Wan Luan Lee, Tsung-Wei Huang, and Yi Zhou, "Enhancing Graph Partitioning with Reinforcement Learning-based Initialization," *IEEE High Performance Extreme Computing (HPEC)*, 2025
  - **Cheng-Hsiang Chiu**, Chedi Morchdi, Chih-Chun Chang, Cunxi Yu, Yi Zhou, and Tsung-Wei Huang, "Optimizing CUDA Graph Scheduling with Reinforcement Learning - A Case Study in SSTA Propagation," *IEEE/ACM International Symposium on Machine Learning for CAD (MLCAD)*, 2025
  - **Cheng-Hsiang Chiu**, Wan-Luan Lee, Boyang Zhang, Yi-Hua Chung, Che Channg, and Tsung-Wei Huang, "A Task-parallel Pipeline Programming Model with Token Dependency," *Workshop on Asynchronous Many-Task Systems and Applications (WAMTA)*, 2025
  - Wan-Luan Lee, Dian-Lun Lin, **Cheng-Hsiang Chiu**, Ulf Schlichtmann, and Tsung-Wei Huang, "HyperG: Multilevel GPU-Accelerated k-way Hypergraph Partitioner," *IEEE/ACM Asia and South Pacific Design Automation Conference (ASP-DAC)*, 2025
  - Boyang Zhang, Che Chang,**Cheng-Hsiang Chiu**, Dian-Lun Lin, Yang Sui, Chih-Chun Chang, Yi-Hua Chung, Wan-Luan Lee, Zizheng Guo, Yibo Lin, and Tsung-Wei Huang, "iTAP: An Incremental Task Graph

Partitioner for Task-parallel Static Timing Analysis," *IEEE/ACM Asia and South Pacific Design Automation Conference (ASP-DAC)*, 2025

- Che Chang, Boyang Zhang, **Cheng-Hsiang Chiu**, Dian-Lun Lin, Yi-Hua Chung, Wan-Luan Lee, Zizheng Guo, Yibo Lin, and Tsung-Wei Huang, "PathGen: An Efficient Parallel Critical Path Generation Algorithm," *IEEE/ACM Asia and South Pacific Design Automation Conference (ASP-DAC)*, 2025
- **Cheng-Hsiang Chiu**, Chedi Morchdi, Yi Zhou, Boyang Zhang, Che Chang, and Tsung-Wei Huang, "Reinforcement Learning-generated Topological Order for Dynamic Task Graph Scheduling", *IEEE High-performance and Extreme Computing Conference (HPEC)*, 2024
- **Cheng-Hsiang Chiu** and Tsung-Wei Huang, "An Experimental Study of Dynamic Task Graph Parallelism for Large-Scale Circuit Analysis Workloads", *IEEE Computer Society Annual Symposium on VLSI (ISVLSI)*, 2024
- Che Chang, **Cheng-Hsiang Chiu**, Boyang Zhang, and Tsung-Wei Huang, "Incremental Critical Path Generation for Dynamic Graphs," *IEEE Computer Society Annual Symposium on VLSI (ISVLSI)*, 2024
- Boyang Zhang, Dian-Lun Lin, Che Chang, **Cheng-Hsiang Chiu**, Bojue Wang, Wan Luan Lee, Chih-Chun Chang, Donghao Fang, and Tsung-Wei Huang, "G-PASTA: GPU Accelerated Partitioning Algorithm for Static Timing Analysis," *ACM/IEEE Design Automation Conference (DAC)*, 2024
- Tsung-Wei Huang, Boyang Zhang, Dian-Lun Lin, and **Cheng-Hsiang Chiu**, "Parallel and Heterogeneous Timing Analysis: Partition, Algorithm, and System," *ACM International Symposium on Physical Design (ISPD)*, 2024
- **Cheng-Hsiang Chiu**, Zhicheng Xiong, Zizheng Guo, Tsung-Wei Huang, and Yibo Lin, "An Efficient Task-parallel Pipeline Programming Framework," *ACM International Conference on High-performance Computing in Asia-Pacific Region (HPC Asia)*, 2024
- Chedi Morchdi, **Cheng-Hsiang Chiu**, Yi Zhou, and Tsung-Wei Huang, "A Resource-efficient Task Scheduling System using Reinforcement Learning," *IEEE/ACM Asia and South Pacific Design Automation Conference (ASP-DAC)*, 2024
- **Cheng-Hsiang Chiu**, Dian-Lun Lin, and Tsung-Wei Huang, "Programming Dynamic Task Parallelism for Heterogeneous EDA Algorithms," *IEEE/ACM International Conference on Computer-aided Design (ICCAD)*, 2023
- **Cheng-Hsiang Chiu** and Tsung-Wei Huang, "Composing Pipeline Parallelism using Taskflow Control Graph," *ACM High-Performance Parallel and Distributed Computing (HPDC)*, 2022
- **Cheng-Hsiang Chiu** and Tsung-Wei Huang, "Efficient Timing Propagation with Simultaneous Structural and Pipeline Parallelisms," *ACM/IEEE Design Automation Conference (DAC)*, 2022
- **Cheng-Hsiang Chiu**, Tsung-Wei Huang, Zizheng Guo, and Yibo Lin, "Pipeflow: An Efficient Task-Parallel Pipeline Programming Framework using Modern C++," https://arxiv.org/abs/2202.00717
- **Cheng-Hsiang Chiu**, Dian-Lun Lin, and Tsung-Wei Huang, "An Experimental Study of SYCL Task Graph Parallelism for Large-Scale Machine Learning Workloads," *International European Conference on Parallel and Distributed Computing (Euro-Par)*, 2021
- Der-Cherng Liaw, **Cheng-Hsiang Chiu**, Chia-Wei Yeh, Chia-Ming Chang, and Hsiao-Jen Hsieh," A Server Load Balancing Design for Peer-To-Peer Network," *International Conference on Computers, Communications, Control and Automation*, 2011
- Der-Cherng Liaw, Chia-Wei Yeh, **Cheng-Hsiang Chiu**, Chia-Ming Chang, and Hsiao-Jen Hsieh, "A Load Balancing Scheme for Web Server Design," *International Conference on System Science and Engineering*, 2011
- Der-Cherng Liaw, Yi-Hung Hsieh, Jing-Hong Lai, and **Cheng-Hsiang Chiu**, "A Network Topology Design for Structural Health Monitoring," *Asian Control Conference*, 2011

- Der-Cherng Liaw, Jing-Hong Lai, **Cheng-Hsiang Chiu**, and Jia-Hong Liao, "A Wireless Sensor Network Platform for Indoor Surveillance System," *International Conference on System Science and Engineering*, 2011
- **Cheng-Hsiang Chiu**, Pang-Chan Hung, Jen-Hui Chuang, and Shing-Lu Huang, "Object Tracking under Sensing Lighting Equipments," *IEEE Conference on Industrial Electronics and Applications*, 2010
- Yi-Yuan Chen, Yuan-Yao Tu, **Cheng-Hsiang Chiu**, and Yong-Sheng Chen, "An Embedded System for Vehicle Surrounding Monitoring," *IEEE Conference on Power Electronics and Intelligent Transportation System*, 2009

Activities

- **Talks**
  - "Pipeflow: C++ Task-Parallel Pipeline Scheduling Framework", CppNow, October 2023
- **Awards**
  - ACM/IEEE DAC Young Student Fellowship, 2021 - 2024
  - ACM/IEEE HPDC Traveling Grant, 2022
  - ACM ISPD Wafer-Scale Physics Modeling Contest - Honorable Mention, 2021
- **Committee**
  - CppCon, 2023 - 2025
  - CppNow, 2023
  - MLCAD, 2024
- **Invited Paper Review**
  - CppCon, 2023 - 2025
  - CppNow, 2023
  - DAC, 2022 - 2023, 2025
  - DATE, 2025
  - Electronics, 2025
  - ICCAD, 2024 - 2025
  - ICCD, 2023
  - Information, 2025
  - IPDPS, 2024
  - ISVLSI, 2024
  - MLCAD, 2024
  - MICRO, 2025
  - SC, 2023
  - Sensors, 2025